

PEFT Efficiency Trade-offs

A structured framework to choose among LoRA, QLoRA, Prefix Tuning, and Adapters

9 min

Duration

B4 – Analyse

Bloom's

L3 – Advanced

Level

AI-FT-IN-TH-000010

ID

PEFT Methods — 4-Dimension Comparison

Dimension	LoRA	QLoRA	Prefix Tuning	Adapters
Trainable params	0.1–4%	0.1–4% (4-bit base)	0.01–1%	0.5–4%
Inference overhead	None (merged)	None (merged)	Context tokens	+5–15% latency
Task performance	★★★★★	★★★★½	★★★★☆☆	★★★★☆
Implementation ease	Easy (PEFT lib)	Moderate (BnB)	Moderate	Moderate (AdapterHub)
Multi-task serving	Multi-adapter	Multi-adapter	Swap prefix	AdapterFusion
Best hardware	1× A10G/4090	1× RTX 3090/4090	Any	Server GPU

Decision Framework — Choose Your PEFT Method

I have ≤ 24 GB VRAM → QLoRA

Load base in 4-bit. Apply LoRA adapters. Train on RTX 3090/4090. Default for most practitioners.

I have 40–80 GB VRAM → LoRA (bf16)

Full-precision base + LoRA adapters. Slightly higher quality than QLoRA. Use A100 40/80 GB.

Constrained generation task → Prefix Tuning

Table-to-text, summarisation, structured output tasks. Best when model input structure is important.

Multi-task BERT-era model → Adapters

Serving many tasks from one base model with AdapterFusion. Ideal for BERT/RoBERTa classification systems.

Need zero inference overhead → LoRA (merged)

Merge adapters into base weights after training. No overhead at inference. Best for production serving.

Maximum quality, sufficient compute → Full FT

Large dataset + A100 cluster. Only when PEFT falls short. Rare in practice.

Multi-Adapter Serving — LoRAX Architecture

Use Case	Without LoRAX / Multi-Adapter	With LoRAX / PEFT Fusion
Serving 10 tasks	10 separate model copies (70× A100 memory)	1 base + 10 small adapters (~1 A100)
Task switching	Reload full model checkpoint per task	Swap adapter in microseconds
New task addition	Train new full model; redeploy	Train new LoRA adapter; add to pool
Memory footprint	$N \times$ full model size	$1 \times$ base + $N \times$ adapter size (~100 MB each)

KEY REFERENCES

- He, J. et al. (2022). Towards a Unified View of Parameter-Efficient Transfer Learning. ICLR 2022.
- LoRAX: huggingface.co/blog/lorax
- HuggingFace PEFT. github.com/huggingface/peft



Next: Concept 12: LoRA Implementation (AI-FT-IN-CD-000002)